Syllabus - Regression Analysis - Fall 2026

Course Information

Course Title: MATH 4780/MSSC 5780: Regression Analysis
Meeting Time: TuTh 12:30pm - 1:45pm
Location: Cudahy Hall 131 (Microsoft Teams - if needed)
Website: http://tinyurl.com/f26-4780

Instructor Details

- **Name:** Mehdi Maadooliat, Ph.D.
- **Office:** CU 351
- **Office Hours:** Tu 1:45pm - 3:15pm & Th 10:45am - 12:15pm in CU 351 or by e-mail

Course Description

This course introduces the theory and application of linear regression for analyzing relationships among variables. Topics include simple and multiple linear regression, statistical inference, model interpretation, diagnostic methods, transformations, categorical predictors, multicollinearity, and model selection. Emphasis is placed on applying regression methods to real data, evaluating model assumptions, and communicating results effectively. The statistical software R will be used for data analysis, visualization, and regression modeling.

Course Topics

- Simple and multiple linear regression
- Least-squares estimation and matrix formulation of regression
- Estimation, inference, and prediction
- Model interpretation and analysis of variance
- Regression diagnostics and residual analysis
- Transformations and remedial measures

- Polynomial, categorical, and interaction effects
- Outliers, leverage, and influential observations
- Multicollinearity, model selection, and model validation
- Regression analysis, visualization, and reporting using R

Learning Outcomes

By the end of the course, students will be able to:

- Fit and interpret simple and multiple linear regression models.
- Apply least-squares estimation and use the matrix formulation of linear regression.
- Perform estimation, hypothesis testing, confidence interval construction, and prediction for regression models.
- Interpret regression coefficients and use analysis of variance to assess regression models.
- Evaluate regression assumptions using residual analysis and diagnostic methods.
- Apply appropriate transformations and remedial measures when model assumptions are not satisfied.
- Model polynomial effects, categorical predictors, and interaction effects.
- Identify and assess outliers, leverage points, and influential observations.
- Diagnose multicollinearity and apply appropriate model-selection and validation techniques.
- Use R to conduct regression analyses, visualize results, and communicate statistical findings effectively.

Additional Learning Outcomes for MSSC 5780

In addition to the learning outcomes above, students enrolled in MSSC 5780 will be expected to:

- Demonstrate a deeper theoretical understanding of regression methods and their assumptions.
- Work with more advanced regression problems requiring greater mathematical and statistical reasoning.
- Critically evaluate alternative modeling strategies and justify model-selection decisions.
- Conduct more extensive independent data analysis and interpretation using R.
- Communicate regression methodology and results at a level appropriate for graduate study.

Prerequisites

- MATH 2780, or MATH 4720
- Some familiarity with R and basic matrix algebra is helpful but not required. Necessary computational and matrix concepts will be introduced as needed.

Textbook

Title: Introduction to Linear Regression Analysis, John Wiley, New York (6th Edition)
Authors: Montgomery, D.C., Peck E.A., and Vining, G.G.
Year: 2021

Grading Breakdown

- **Homework**: 30%
- **In-class Exam**: 30%
- **Final Project**: 30%
- **Best of three**: 10%

Students enrolled in MSSC 5780 will complete additional or more advanced work on selected homework assignments, the in-class exam, and/or the final project. These additional requirements are designed to assess the graduate-level learning outcomes described above.

You will **NOT** be allowed any extra credit projects to compensate for a poor average. The final grade is based on the following scale:

Grading Scale

Grade	Range
A	93.5 - 100
A-	90.0 - 93.5
B+	86.5 - 90.0
B	83.5 - 86.5
B-	80.0 - 83.5
C+	76.5 - 80.0
C	73.5 - 76.5
C-	70.0 - 73.5
D+	66.5 - 70.0
D	63.5 - 66.5
D-	60.0 - 63.5

Grade	Range
F	below 60.0

Note: No late homework will be accepted, and no make-up work will be allowed unless in the case of a university-recognized excused absence/documented emergency. Submit incomplete work if necessary, but ensure scanned PDFs are legible before submission.

Exam Dates

- **In class Exam**: Nov 10th (70 minutes long)
- **Final Project**: Dec 15th from 10:30 am to 12:30 pm

Homework Policy

- Homework is required for a better understanding of the material and must be submitted to D2L.
- No late homework will be accepted, and the lowest homework assignment grade will be dropped. It is recommended to submit even incomplete work.
- You can scan your homework and submit it as a PDF, but ensure it is readable.
- For data-analysis problems, relevant R code and output should be included with the submitted work. Computational results should be clearly organized and accompanied by appropriate interpretation.

Generative AI

Generative AI tools such as ChatGPT may be used for clarification, coding assistance, and take-home work unless otherwise stated. Students remain fully responsible for the correctness and interpretation of all submitted work and must disclose substantive use of generative AI. Generative AI is not permitted during the in-class exam.

Attendance Policy

Attendance is required, and the College of Arts and Sciences attendance policy will be observed.

Make-up Policy

There will be no make-up exams or homework unless there is a university-recognized excused absence/documented emergency

Honor Code

Marquette University Honor Code:
"I recognize the importance of personal integrity in all aspects of life and work. I commit myself to truthfulness, honor, and responsibility, by which I earn the respect of others. I support the development of good character and commit myself to uphold the highest standards of academic integrity."

Important dates

- **Monday, August 31:** Classes begin
- **Tuesday, September 8:** Drop/add ends
- **Thursday - Friday, October 22 - 23**: Fall Break
- **Friday, November 20:** Last day to withdraw with W
- **Wednesday - Sunday, November 25 - 29**: Thanksgiving Break
- **Saturday, December 12**: Classes end
- **Tuesday, December 15th, 10:30 am - 12:30 pm**: Final Project

For more important dates, see the full MU Academic Calendar.

Important Note

The syllabus may be modified throughout the course. Any substantial modifications will result in a reissued syllabus.